

Performance Work Statement (PWS)
Fully functional, military-ready smartphone application for Compromised Driving Detection for Warfighter Readiness

GENERAL INFORMATION

1.0 Scope of Work: Contractor will design, build, and deliver a complete mobile-based compromised driving detection system that can be deployed across motorcycle-riding service members. This includes software development, AI-driven behavior detection algorithms, on-device real-time processing, user feedback features, and backend support infrastructure — all aimed at driving measurable improvements in safety and readiness through a smartphone application for compromised driving (“SACD”).

The goal of this initiative is to study the viability of a new technology for smartphone that will reduce avoidable driver behavioral errors caused by fatigue – thereby saving injuries, lives, trauma, property, costs, and maintaining operational impact – without the cost or problems associated with additional hardware/gear and to deliver a SACD which is ready for deployment.

1.1. Key Components:

- 1.1.1 Develop and deliver a real-time fatigue and behavior-based motorcycle driving detection application (SACD).
- 1.1.2 Package the SACD with a secure backend architecture for optional data analysis and policy support.
- 1.1.3 Ensure full functionality on both Android and iOS platforms.
- 1.1.4 Deploy and evaluate the system with a representative service member group, incorporating feedback into the final product.
- 1.1.5 Provide full transition documentation and training support for large-scale military deployment.

2.0 Background:

This initiative is in response to the record-level of motorcycle incidents and fatalities within the Department of War (DOW) that have occurred this year, as well as to the report on service member fatigue by DOW and the U.S. Government Accountability Office (GAO). Since the report was released in March 2024, DOW has faced increasing external pressures, including Congress, to address the problem of fatigue.

U.S. Government research shows that the rider is at fault in 73% of multi-vehicle incidents and that behavior-based errors, including fatigue, indiscipline, excessive speed and alcohol use are the leading causal factors in these mishaps.

Motorcycle-related incidents, particularly those linked to fatigue and behavioral risk factors, continue to degrade warfighter readiness and result in preventable fatalities across the DOW.

CONTRACTOR REQUIREMENTS

3.0 Technical Requirements/Tasks for the development of SACD:

3.1 Sensor Integration and Processing

- 3.1.1 Utilize existing smartphone sensors (accelerometer, gyroscope, GPS, magnetometer, barometer and others as they become available by the phone manufacturers) for behavioral detection.
- 3.1.2 Integrate proprietary signal filtering and sensor fusion technology to ensure high accuracy and minimize false alerts.

3.2 On-Device AI Engine

- 3.2.1 Embed machine learning algorithms capable of classifying fatigue and risk-level driving behaviors.
- 3.2.2 Algorithms will be designed for on-device processing to protect user privacy and ensure real-time intervention.

3.3 SACD Mobile Application Design

- 3.3.1 Cross-platform mobile application with intuitive UI/UX for non-distracting use during rides.
- 3.3.2 Real-time feedback mechanisms including alerts, behavior summaries, and rest suggestions.
- 3.3.3 Built-in mechanisms for user engagement and potential gamification (if desired by DOW).

3.4 Simulation and Testing

- 3.4.1 Simulate a range of high-risk riding behaviors and fatigue indicators using controlled environments and digital modeling.

3.4.2 Gather real-world data through Contractor's own tools for algorithm validation and refinement.

3.5 Deployment and Integration

3.5.1 Conduct pilot deployment with Navy motorcycle riders to ensure real-world functionality.

3.5.2 Implement continuous integration and updates based on live user data and feedback.

3.5.3 Ensure all system components are production-ready and scalable.

4.0 Government Furnished:

4.1 No Government Furnished Equipment will be required for the execution of this project. Contractor will utilize its own commercially available tools, development environments, and simulation resources.

4.2 Notwithstanding, the Navy will provide access to volunteer Navy motorcycle riders and their personal devices after consent is given by the sailors.

4.3 All required hardware, software, and data acquisition platforms outside of access to the voluntary riders' smartphones will be contractor-furnished.

5.0 Deliverables / Schedule:

5.1 At the conclusion of this project, Contractor will deliver a **fully functional, military-ready smartphone application (SACD)**. This deployment-ready application will be capable for use across all service branches, requiring no additional hardware or wearables.

5.1.1 All major system components — including the app (iOS/Android), behavior detection algorithms, backend services, and user feedback tools as jointly developed in this project — will be provided in a hardened, scalable form.

5.1.2 Training materials and support documentation for the deployment-ready SACD app will also be included.

5.2 **Tasks and Associated Deliverables:** Timely submission of deliverables is essential to successfully completing this requirement.

5.2.1 Schedules for deliverables are specified in Exhibit A.

5.2.2 All deliverables shall be prepared and submitted according to format, content, and schedule described in the PWS.

5.3 **Reporting Requirements:** The contractor is required to provide written progress reports monthly until final product is delivered. The progress report shall identify:

5.3.1 All work completed during the specified period and shall present the work to be accomplished during the subsequent period.

5.3.2 Any problems that arose and a statement explaining how the problem was resolved.

5.3.3 Any problems that have arisen but have not been completely resolved and provide an explanation.

EXHIBIT A

Timeline of Deliverables	
Deliverable	DUE
MOTORCYCLE ACCIDENT SIMULATION	MONTH 3
FATIGUE SIMULATOR	MONTH 9
DATA COLLECTION & ANALYSIS	MONTH 12
ALGORITHM DEVELOPMENT AND IMPROVEMENTS	MONTH 12
POC APPLICATION DESIGN	MONTH 12
PILOT DEVELOPMENT (APP DEVELOPMENT AND CONTINUOUS IMPROVEMENTS)	MONTH 15
APP HARDENING FOR GTM PRODUCT	MONTH 17
GTM PROJECT PLANNING & APPROVAL	MONTH 20
FINAL PRODUCT	MONTH 22

6.0 **SACD Application Deployment:** For sake of clarity, deployment support, and continuing licensing, service, and maintenance of the SACD application for services branches and members is outside of the scope of this SACD development project and will be contemplated via a separate agreement.

7.0 Travel:

7.1 Travel, included within this agreement, will be limited to locations where the pilot with Navy riders will be conducted to facilitate smooth SADC development.

7.2 Outside of this need, no travel is required or anticipated for the successful execution of this project.

8.0 Contractor's Key Personnel:

Key positions, already available via Contractor personnel, are:

- *Lead compromised driving Scientist*
- *Lead mobile AI Engineer*
- *Lead Data Scientist*
- *UX/UI and User engagement Expert*
- *Lead Mobile Engineer*
- *Lead Software Engineer*
- *Experienced mobile deployment Manager*
- *Experienced Program Management*

9.0 Security Requirements:

Contractor complies with the most stringent data privacy laws in the US and follows GDPR guidelines. At this point no clearance level is required for the performance of this contract.

10.0 Data Rights:

The data collected in this project consists of driving behavioral data through the personal smartphones of the Navy motorcycle riders and is owned by them and/or the United States Government. Contractor will have the right to use an anonymized form of the data to develop and improve the work product of this project. All data is stored in compliance with the CCPA (California Consumer Privacy Act) and the GDPR standard of the European Union.

11.0 Section 508 – Electronic and Information Technology Standards

All electronic and information technology (EIT) deliverables provided under this contract comply with the accessibility standards of Section 508 of the Rehabilitation Act (29 U.S.C. §794d), as amended.